

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B. Tech. Branch : Computer Science & Engineering / Computer Engineering/Computer Science & Engineering (AI& ML)/ AIDS (2020-2021)/ Computer Science & Design Semester : IV Subject Code & Name: BTES405 Digital Logic Design & Microprocessor Max Marks: 60 Date:25-01-24 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain error detecting and correcting codes.	Understand	6
B)	Define digital signals. Explain positive logic and negative logic.	Understand	6
C)	Explain types of Boolean postulates with example.	Understand	6
Q.2	Solve Any Two of the following.		12
A)	Implement the expression using a multiplexer. $F(A,B,C,D) = \sum(0,2,3,6,8,9,12,14).$	Analysis	6
B)	Design half adder and full adder circuit.	Synthesis	6
C)	$F(A,B,C,D) = \pi M(1,2,3,8,9,10,11,14).d(7,15).$	Analysis	6
Q. 3	Solve Any Two of the following.		12
A)	How do you eliminate the race around condition in a J-K Flip-Flop.	Remember	6
B)	What is meant by programmable counter? Mention its application.	Remember	6
C)	Write a short note on Parity Generator/ Checker.	Understand	6
Q.4	Solve Any Two of the following.		12
A)	What are the various registers of 8085 microprocessor. Discuss their function.	Remember	6
B)	Draw the block diagram of 8086 and explain its each block in brief.	Understand	6
C)	Explain different addressing modes of 8086.	Understand	6
Q. 5	Solve Any Two of the following.		12
A)	Explain DMA Controller with neat block diagram.	Understand	6
B)	Explain following instructions 1) CMP. 2) AAS. 3) IMUL. 4) CBW. 5) TEST. 6) RCR.	Analysis	6
C)	Explain with examples different types of arithmetic instructions.	Analysis	6
*** End ***			